

Sentiment-Augmented Quarterly Stock Prediction

A leakage-aware supervised learning pipeline for predicting 63-day sector-relative stock performance after quarterly filing events

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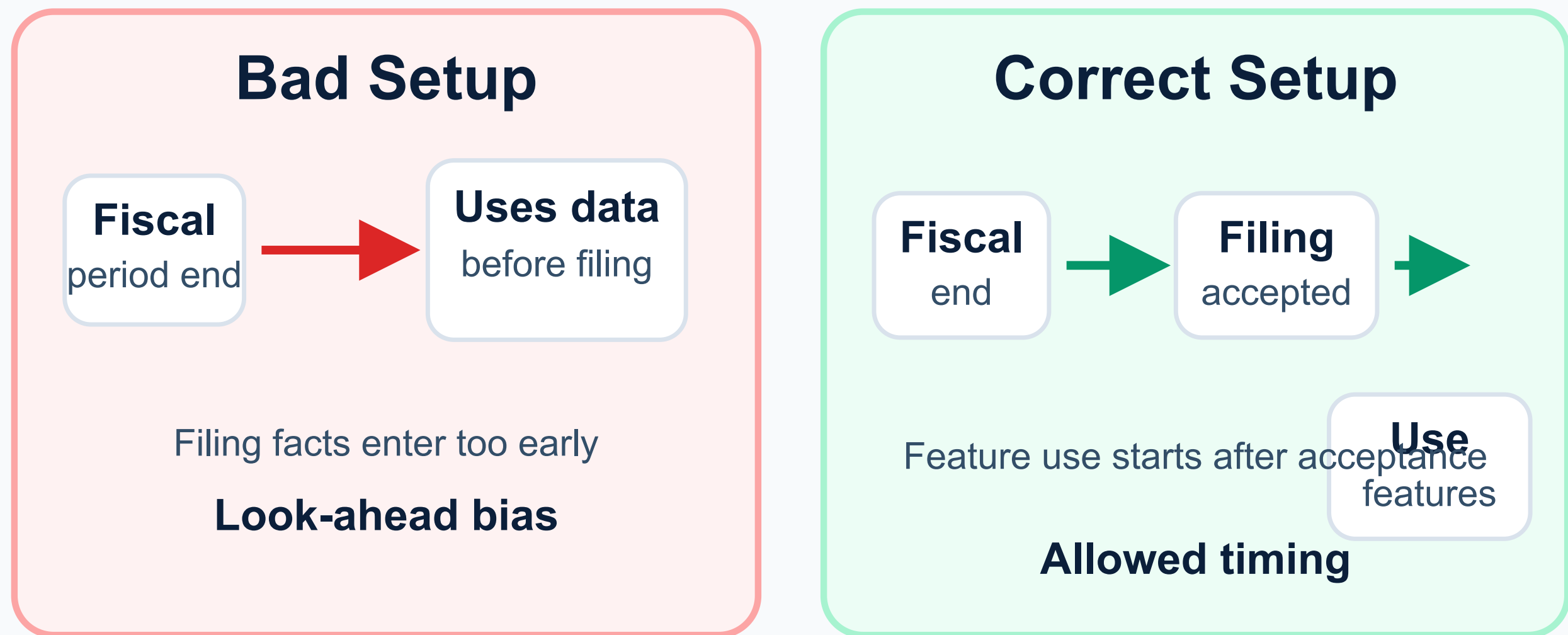
This project tests whether accounting fundamentals, market context, and sentiment/event-specific features can predict whether a stock will outperform its sector after a quarterly filing event.

Why This Problem Is Difficult

Financial prediction is noisy, time-dependent, and vulnerable to look-ahead bias. A model can appear predictive if it accidentally uses information that would not have been available at the time of prediction. The project therefore focuses on building a fair event-level experiment, not simply maximizing accuracy.

- No random train/test splits
- No future information in features
- No post-filing facts before they become tradable
- Labels must not overlap validation windows

Leakage Control



Research Question

Can a machine learning model predict whether a company will outperform its sector over the next 63 trading days after a quarterly filing event?

Sub-question: Do sentiment and event-specific features add predictive value beyond fundamentals and market context?

- Fundamentals provide a weak but useful baseline.
- Market context may improve event timing.
- Event-specific sentiment may add incremental signal.

Data Sources

Source	Role	Examples
SEC EDGAR	Fundamentals and filings	10-Q/10-K facts, filing text
Market prices	Returns and volatility	Pre-event returns, volatility, volume
Alpha Vantage	Earnings metadata	EPS surprise, revisions, analyst count
Capital IQ / universe data	Company context	Sector, ticker universe
Sentiment model	Text features	Positive, negative, neutral probabilities

Conclusion

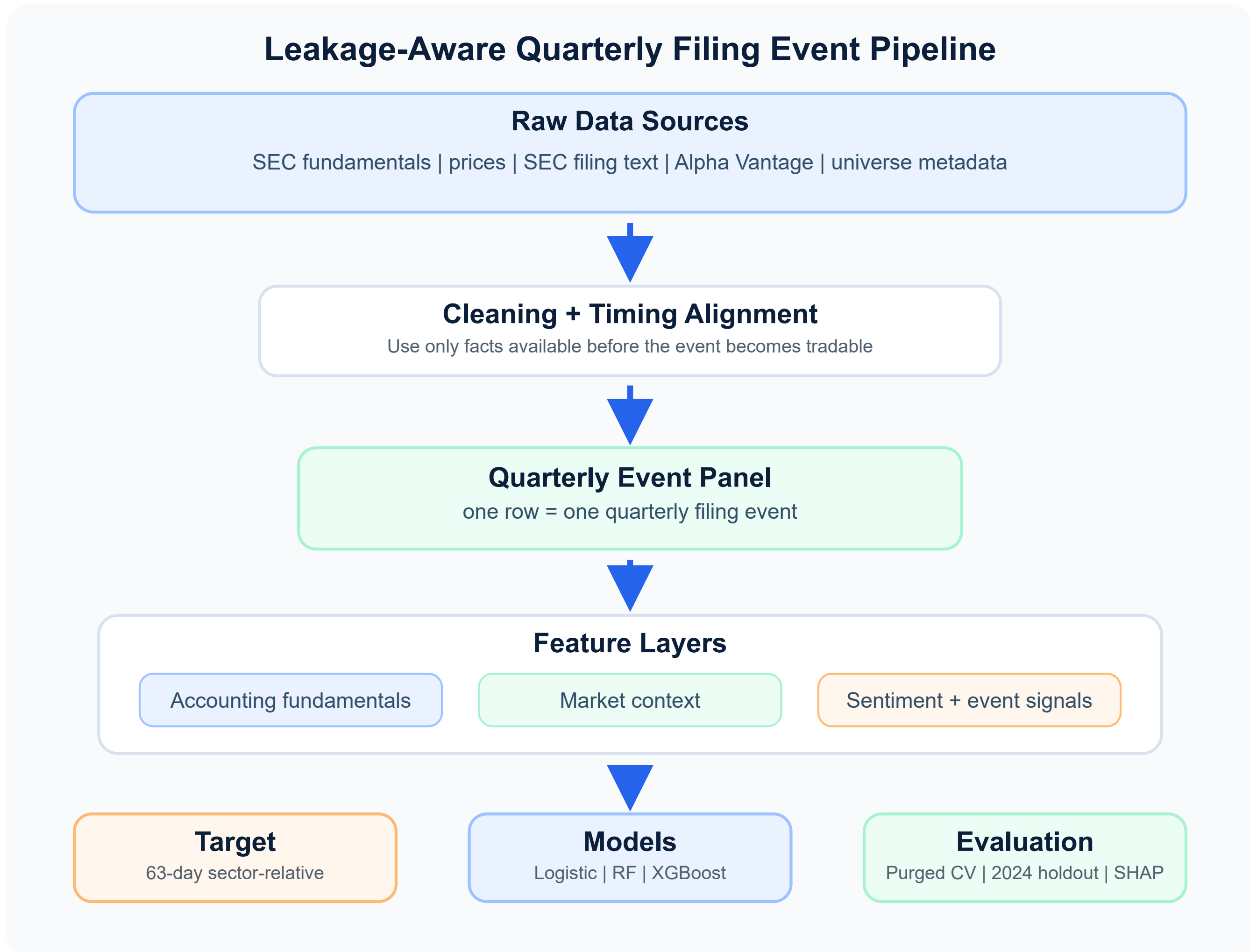
The project does not claim to be a finished trading strategy. It demonstrates a leakage-aware research framework for testing whether financial signals add predictive value around quarterly filing events.

Limitations and Future Work

- Predictive signal is modest and unstable.
- Public financial data requires careful timing alignment.
- SEC filing text can be diluted by boilerplate language.
- Some promising results are from legacy 21-day experiments, not the final 63-day contract.

Next: complete the final 63d rerun, add higher-quality market/accounting data, expand event-specific sentiment, and move toward portfolio construction only after signal stability improves.

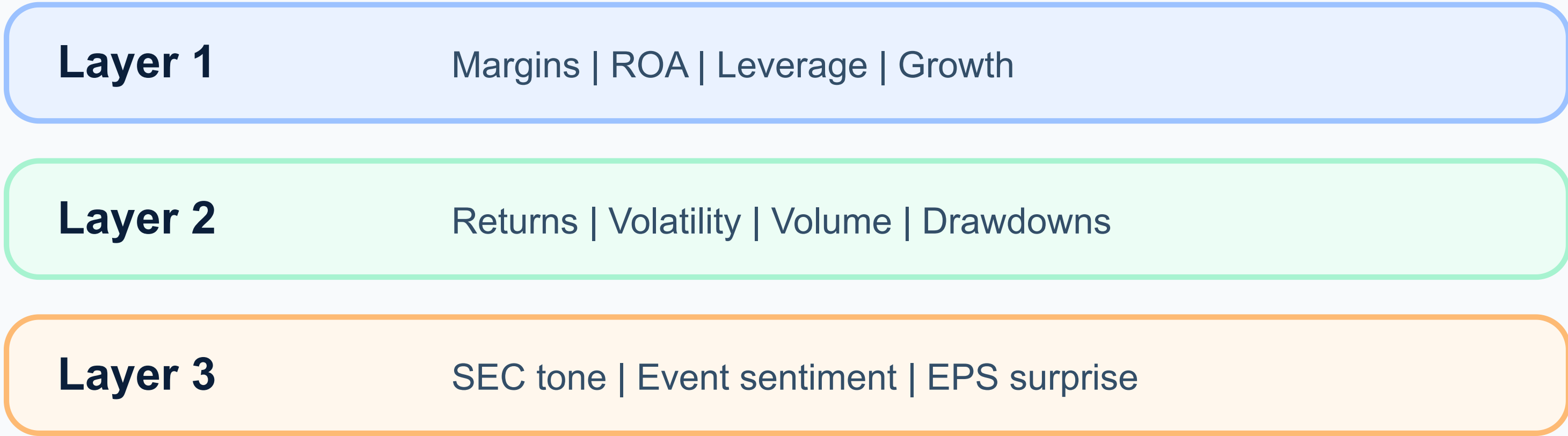
Modeling Pipeline



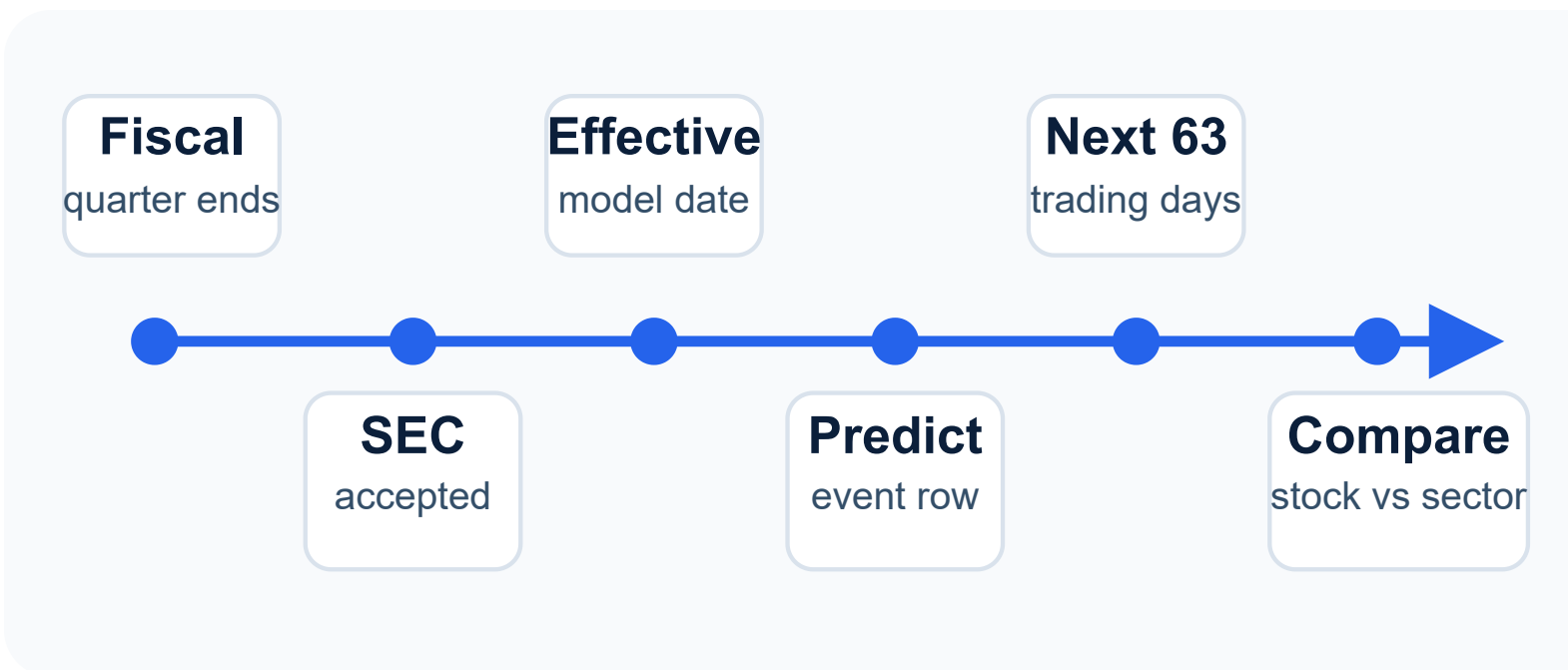
Each prediction is made at the filing-event level using only information available before the event becomes tradable.

Feature Design

Controlled Feature-Addition Experiment



Event Timing



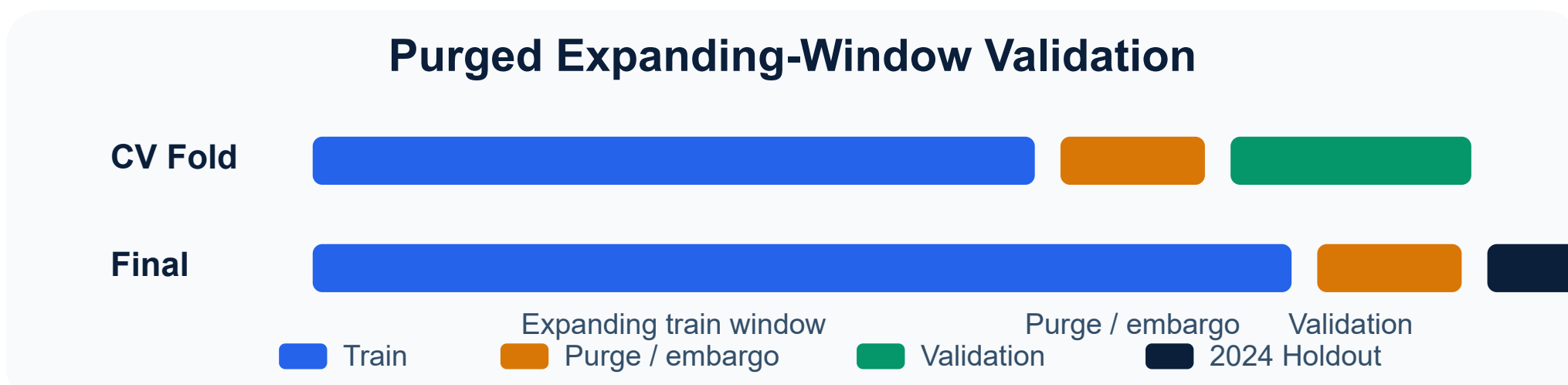
Target

1 if stock outperforms its sector benchmark over the next 63 trading days
0 otherwise

Algorithms Compared

Logistic Regression Linear baseline
Random Forest Nonlinear ensemble
XGBoost Boosted-tree model

Target and Validation

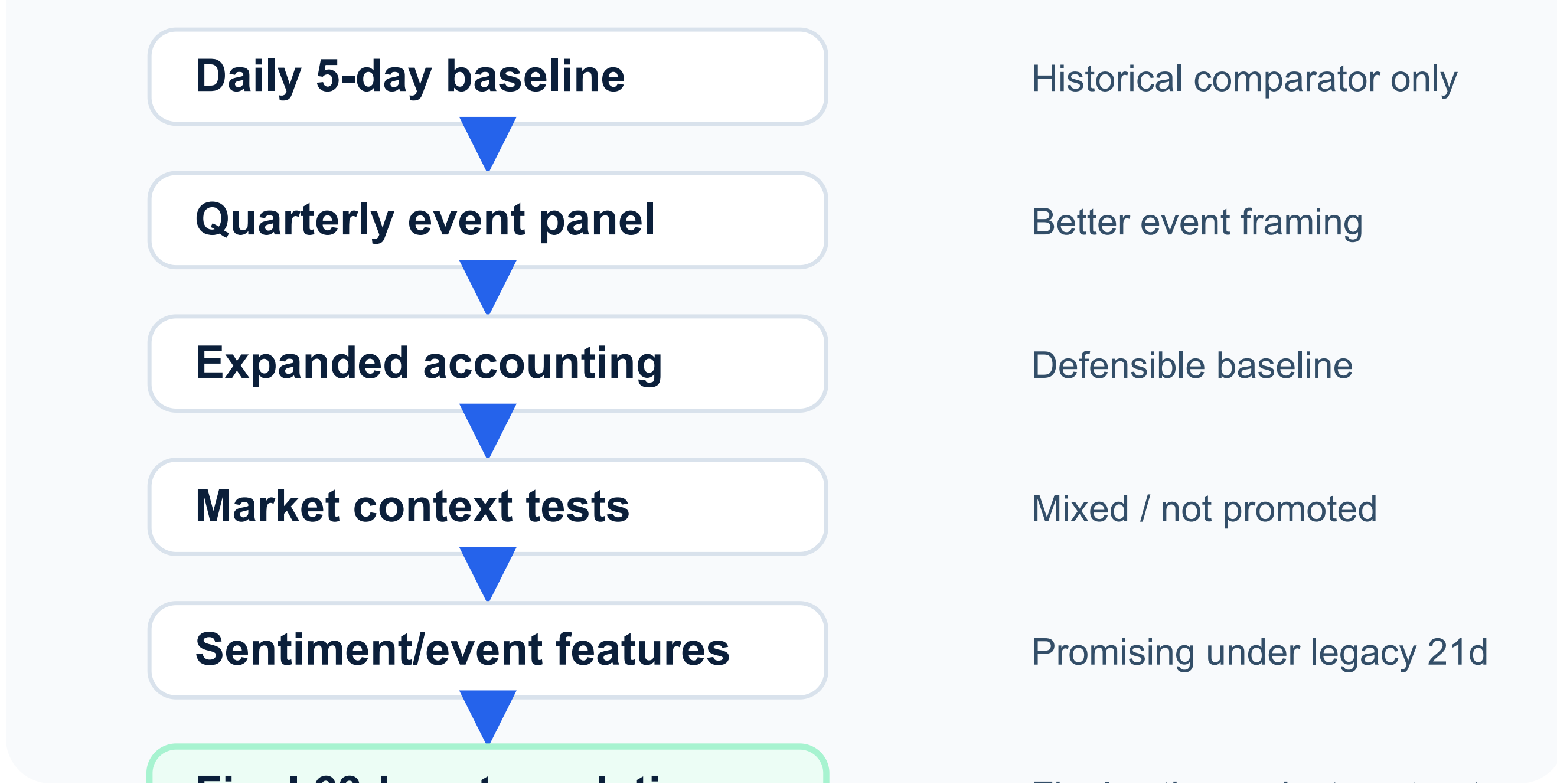


Why it matters: Because the label uses future returns, nearby observations can overlap with validation outcomes. The purge/embargo removes contaminated training rows.

5-fold purged expanding-window cross-validation, fixed 2024 holdout, 5-day embargo, and no random shuffling.

What the Experiments Showed

Project Evolution and Results Framing



The strongest checked-in sentiment result comes from a legacy 21-day benchmark. The final project contract uses the stricter 63-day sector-relative target.

Selected Benchmark Evidence

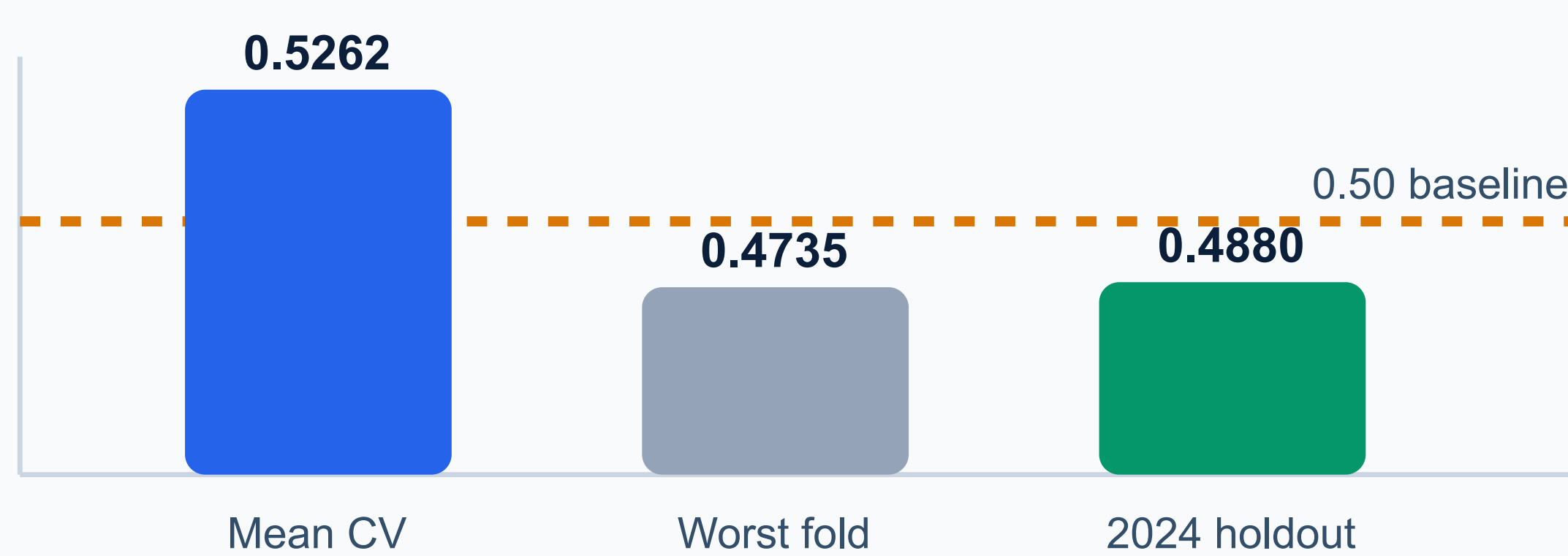
63d Stability Candidate

Selected model: XGBoost
Mean CV AUC: 0.5262
Worst-fold AUC: 0.4735
2024 holdout AUC: 0.4880

Legacy 21d Sentiment Test

Winner: event-specific sentiment only
Model: Random Forest
Holdout AUC: 0.5367
Not the final 63d label contract

63d Stability Candidate AUC



Modest CV signal but weaker holdout performance; not a trading-ready edge.

Sentiment Feature Comparison

No sentiment Core controls	Broad filing SEC text	Event-specific Random Forest AUC 0.5367	Combined Mixed signal
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Legacy 21d benchmark, not final 63d contract

Event-specific sentiment looked more useful than broad filing sentiment under the legacy label.

Main Findings

- Quarterly filing events are a better modeling unit than daily stock rows.
- Leakage control is central to the project, not a minor detail.
- Fundamentals created a defensible baseline, but predictive signal was weak.
- Generic market features did not consistently improve performance.
- Event-specific sentiment was more promising than broad filing sentiment.

Next Steps

Final claims should be based on completed 63d sector-relative benchmark artifacts, stability checks, and reproducible configs.

Direction: better data, broader event-specific sentiment coverage, stronger diagnostics, and portfolio construction only after signal stability improves.